defined at runtime. This allows for a page to have a completely new template for the user every time he visits, leading to a much richer app-like feel.

Building your own service and using it:

Let's now create our own service and use it in an angular application. The purpose of a service can vary a great deal, from simple computation to data fetching, extraction and even complex manipulations. Lets try to build a service that randomly (pseudo - random technically) gives us a 'heads' or a 'tails' String. Services, like most other angular components are also classes, hence we start by creating a class as follows:

- Create a new file under the app folder in your working application. (in case you haven't created one already, please follow the steps listed under "Working application" from Basics chapter to current)
- Name it as toss.service.ts

One of the conventions for naming services are

```
<module _ name _ dash _ seperated> _ <service _ name>.ser-
vice.ts
```

Add the following lines in it

```
@Injectable()
export class TossService {}
```

Notice the injectable annotation (do remember the parenthesis) that notifies angular of possible Dependency Injections.

Next we add a function coinFlip that returns a string with value 'heads' or 'tails' randomly.

```
@Injectable()
export class TossService {

function coinFlip() {
  return (Math.floor(Math.random() * 2) == 0) ? 'heads' :
'tails';}}
```

The above function coinFlip is a very basic implementation where a number between 0 and 1 is chosen with random probability and if the number chosen (n) is greater than 0.5 then we show tails, else we show heads.

```
N -> Random Number
M = Floor value of  [N*2] (can either be 1 or 0.)
```